

Section 04



Industrial Security (Bachelor)

Section 04 – Industrial Protocols

Industrial Protocols

Lecture Notes

Department of Computer Science

- 1 Legacy Heritage
- 2 Modbus
- 3 Siemens, DNP3, EtherNet/IP
- 4 Modern Secure Protocols

By the end of this section you will be able to

- Recognise common industrial protocols and their use cases.
- Understand why legacy protocols typically lack authentication.
- Decode a Modbus TCP frame by hand.
- Compare insecure legacy protocols with secure modern alternatives (OPC UA, MQTT-TLS).
- Identify when secure variants are appropriate.

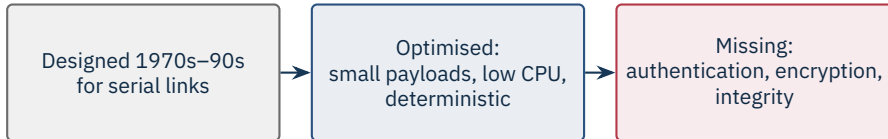
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Legacy Heritage

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Core fact

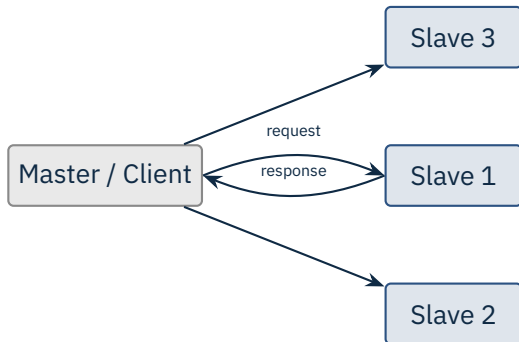
Most industrial protocols in production today were **designed before the Internet** and provide no authentication, integrity, or confidentiality.



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Modbus

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Variants: Modbus RTU (serial), Modbus TCP (port 502)
Address space: coils, discrete inputs, input regs, holding regs
No auth, no encryption – by design (1979)

Source: Modbus Organization, Modbus Application Protocol Specification v1.1b3, 2012.



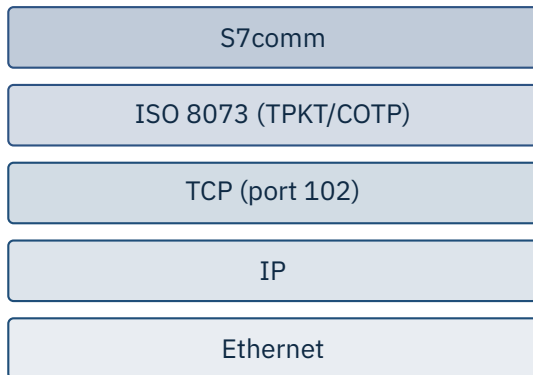
1 00 01 00 00 00 06 01 03 00 00 00 0A

2 ^TID^ ^PID^ ^LEN^ ^U^ ^FC^ ^Addr^ ^Cnt^ -> Read 10 Holding Regs from 0x0000

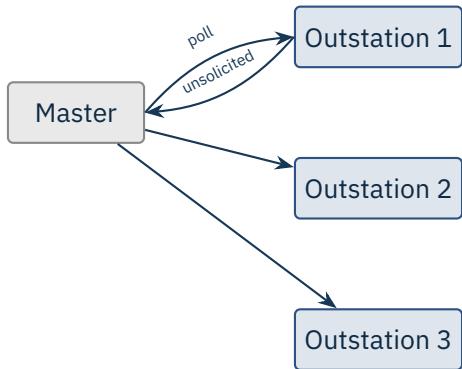
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Siemens, DNP3, EtherNet/IP

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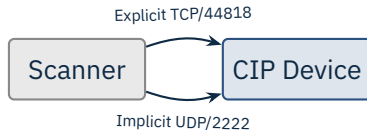


Read/write variables, upload/download programs, start/stop CPU. S7commPlus added session keys
– still broken in practice.



DNP3 dominant in NA utilities; IEC 60870-5-104 in Europe. Both ride TCP today; both added Secure Authentication (SAv5) on top, but encryption is optional.

Source: IEEE 1815-2012 (DNP3) | IEC 60870-5-104:2006 + Amd. 1:2016.



CIP carries object-oriented messages (assemblies, parameters, identity).
CIP Security adds TLS/DTLS and certificate-based authentication.

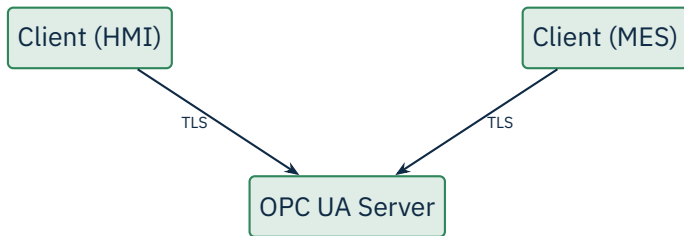


EtherType 0x8892 – bypasses IP for real time.
IRT needs hardware switching for sub-ms cycles.

4

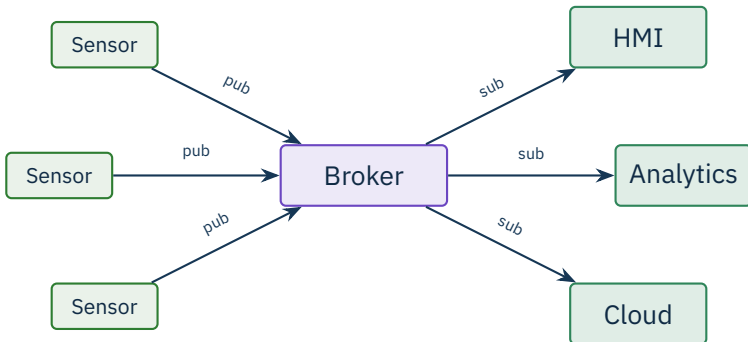
Modern Secure Protocols

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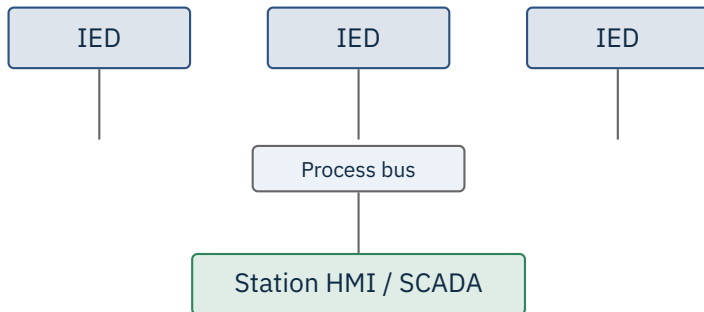
X.509 mutual auth • message signing • encryption
Rich information model • pub/sub since v1.04

Source: IEC 62541 (multipart, OPC Foundation) | OPC UA Specification Part 2 (Security Model).



Default 1883 plaintext / 8883 with TLS. Sparkplug B adds OT semantics.

Protocol	Auth	Integrity	Confid.	Notes
Modbus	×	×	×	1979, ubiquitous Siemens-proprietary Utilities
S7comm	×	×	×	
DNP3	opt SAV5	opt	×	
EtherNet/IP	opt CIP	opt	opt	Rockwell
Profinet	weak	×	×	Siemens / IRT
OPC UA	X.509	sign	TLS	modern default
MQTT	+TLS	TLS	TLS	IIoT messaging



The substation stack

MMS for client/server; **GOOSE** for sub-4 ms trip messages; **Sampled Values** for measurements. Goose has *no* authentication by default.

Source: IEC 61850 (multipart) | IEC 62351 (security for IEC TC 57 protocols).

★ Key Takeaway: What to remember from this section

- Most legacy industrial protocols are unauthenticated and unencrypted by design.
- Modbus is the canonical example – 12 bytes of MBAP and a one-byte function code.
- Modern protocols (OPC UA, MQTT-TLS) bring real cryptography, but require correct configuration.
- Network-layer mitigations (firewalls, segmentation) compensate where the protocol cannot.

- Modbus Organization. *Modbus Application Protocol Specification* v1.1b3, 2012.
- Modbus Organization. *Modbus/TCP Security*, 2018.
- IEEE 1815-2012. *Electric Power Systems Communications – DNP3*.
- IEC 60870-5-104:2006 + Amd. 1:2016. *Telecontrol equipment – Network access for IEC 60870-5-101*.
- IEC 62541 (multipart). *OPC Unified Architecture* (OPC Foundation).
- IEC 61850 (multipart). *Communication networks and systems for power utility automation*.
- IEC 62351 (multipart). *Security for TC 57 protocols*.
- OASIS. *MQTT Version 5.0*, 2019.
- ODVA. *The CIP Networks Library, Volume 1-8*, CIP Security spec.
- Siemens. *Step7 & S7 communication protocols* (S7comm, S7commPlus).
- Wightman, R. *Project Basecamp – ICS Vulnerability Catalog*, Digital Bond, 2012.

End of Section 04



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Questions and discussion